Assignment 4 – Procurepro

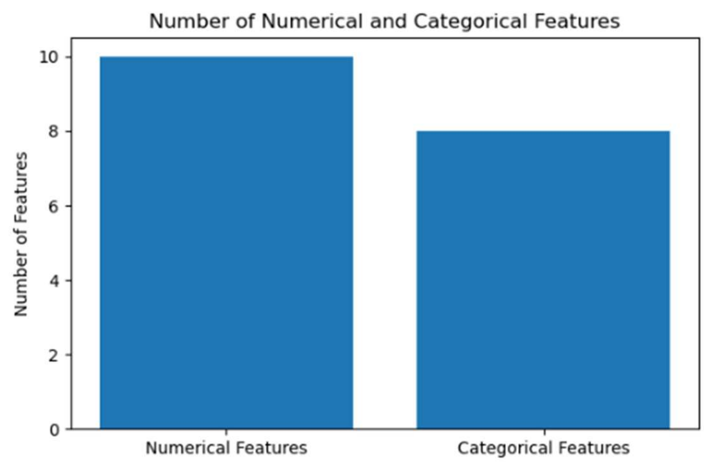
Section A		
#	**Questions**	**Answers**
1	Confusion matrix	True Negative: 18 False Positive: 11 False Negative: 12 True Positive: 31
2	Accuracy	68.06%
3	Precision	73.81%
4	Recall	72.09%
5	F1-score	72.94%
6	Training Accuracy	86.81%
7	Testing Accuracy	68.06%

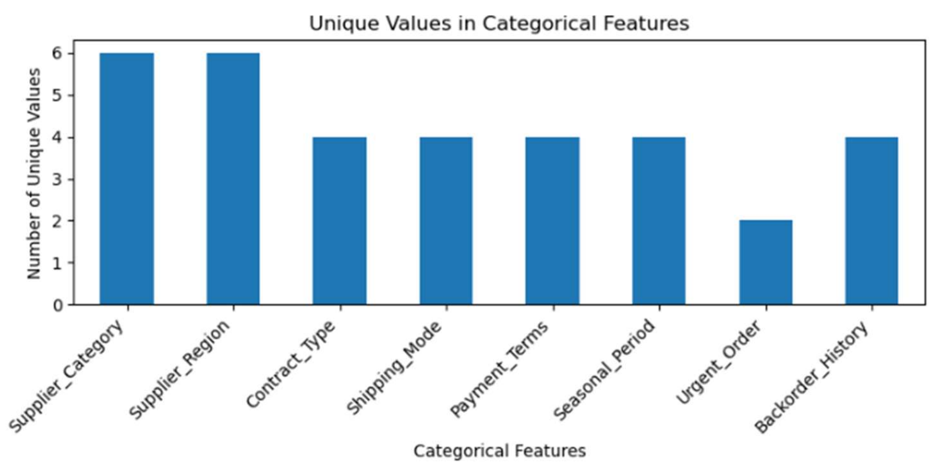
Section B		
#	**Questions**	**Answers**
1	What is the business about?	This is a B2B procurement service that helps medium-sized organizations source, order, and manage office-related products
2	What problem is the business trying to solve?	The main business problem is supplier delivery uncertainty
3	What decision can machine learning help the business make?	Machine learning can help the business decide which purchase orders require early attention. In this assignment, students will train a Decision Tree classification model to predict whether a purchase order has high supplier delay risk.
4	What is the target variable in the dataset?	Supplier_Delay_Risk
5	What are the input features?	Every column except Purchase_Order_ID and Supplier_Delay_Risk
6	Why is this prediction useful for the business?	This prediction is useful because procurement teams have limited time.

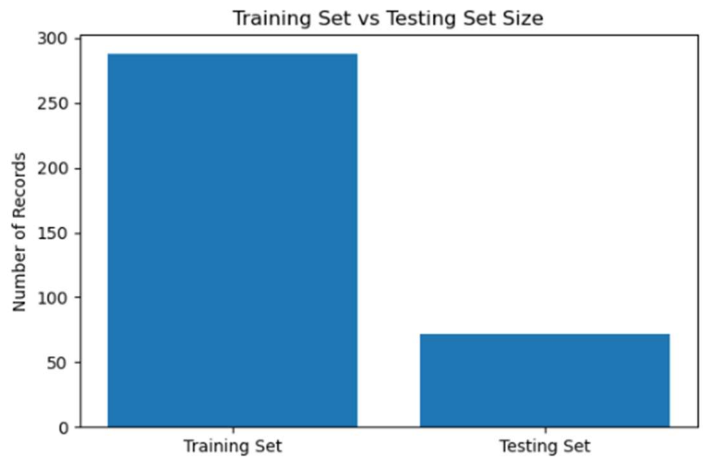
Section C		
#	**Questions**	**Answers**
1	Then briefly explain whether the Decision Tree model may be overfitting	Yes, the Decision Tree model shows some signs of overfitting. The controlled Decision Tree achieved a training accuracy of **86.81%** and a testing accuracy of **68.06%**, resulting in a difference of approximately **18.75 percentage points**. This indicates that the model performs significantly better on the training data than on unseen testing data. The overfitted Decision Tree model further confirms this observation. It achieved **100% training accuracy** but maintained the same testing accuracy of **68.06%**. This means the model memorized the training data rather than learning patterns that generalize well to new purchase orders. Therefore, limiting the tree depth helped reduce overfitting and produced a more interpretable model.
2	How well did the Decision Tree model perform?	The Decision Tree model performed reasonably well. It achieved a testing accuracy of **68.06%**, a precision of **73.81%**, a recall of **72.09%**, and an F1-score of **72.94%**. The model was able to correctly identify many high-risk supplier orders while maintaining a balance between precision and recall.

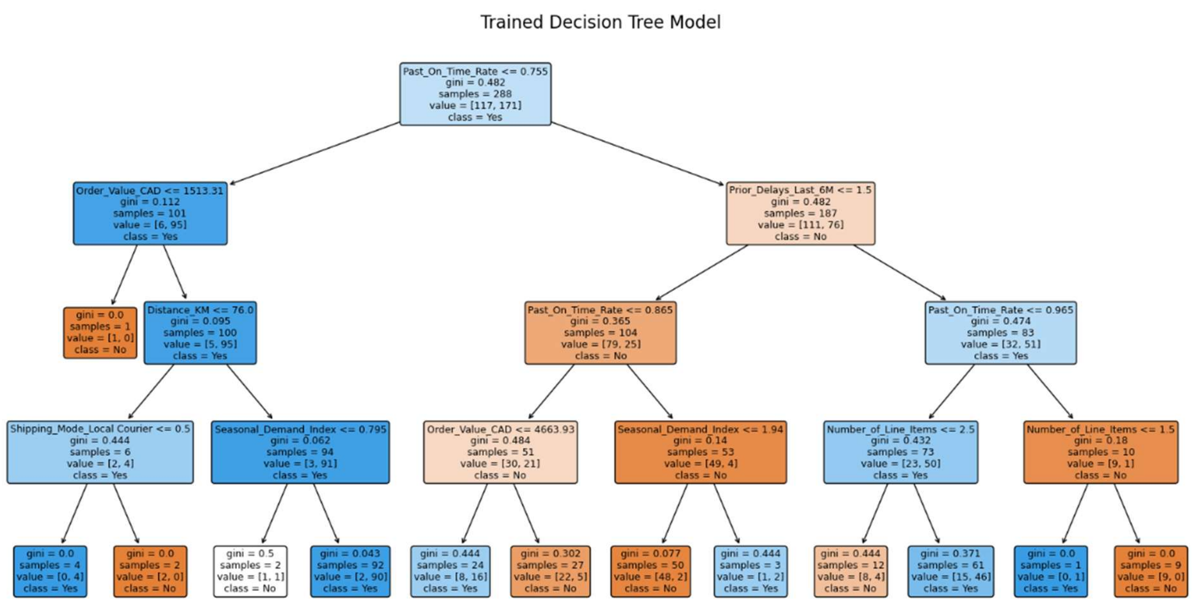
3	Is there a large difference between training accuracy and testing accuracy?	Yes. The training accuracy was **86.81%**, while the testing accuracy was **68.06%**. The difference of approximately **18.75 percentage points** suggests that the model learned the training data better than it generalized to unseen data.
4	Does the model show signs of overfitting? Why or why not?	Yes. The gap between training and testing accuracy indicates some overfitting. This is further supported by the overfitted Decision Tree model, which achieved **100% training accuracy** but did not improve testing accuracy.
5	Which evaluation metric is most important for this business problem?	**Recall** is arguably the most important metric for this business problem. Recall measures how many actual high-risk supplier orders were correctly identified by the model.
6	What do false positives mean in this business context?	A false positive occurs when the model predicts that a purchase order is at high risk of supplier delay, but the order actually arrives on time. In this situation, procurement staff may spend extra time monitoring or following up on an order that does not require intervention. While this may create additional workload, the business impact is generally lower than missing an actual high-risk order.
7	What do false negatives mean in this business context?	A false negative occurs when the model predicts that a purchase order is low risk, but the supplier delivery is actually delayed. This is more serious because procurement teams may fail to take preventive actions. False negatives can lead to inventory shortages, project delays, emergency shipping costs, and reduced customer satisfaction.
8	Which features were most important in the Decision Tree model?	Based on the business problem and the structure of the Decision Tree, the most influential features are likely to include: • Prior_Delays_Last_6M • Past_On_Time_Rate • Backorder_History • Inventory_Buffer_Days • Supplier_Rating • Promised_Lead_Time_Days These features directly relate to supplier reliability, historical performance, and operational risk.
9	How can these important features help the business make better decisions?	These features provide valuable insight into supplier performance and delivery risk. For example: • Suppliers with many prior delays may require closer monitoring. • Low on-time rates may indicate unreliable suppliers. • Frequent backorders may signal supply chain instability. • Low inventory buffers may increase the impact of delays. • Poor supplier ratings may indicate broader performance issues. Procurement managers can use this information to select alternative suppliers, increase inventory buffers, expedite shipping, or proactively communicate with customers.
10	What is one possible limitation or bias in the model or dataset?	One limitation is that the dataset is synthetic and may not fully represent real-world supplier behavior. Historical data may also not capture external factors such as weather disruptions, labor strikes, transportation issues, or sudden market demand changes. These factors could affect supplier delivery performance but are not included in the model.
11	Why should human judgment still be used when making business decisions based on model results?	Human judgment is important because machine learning models cannot capture every business circumstance. Procurement managers may have access to additional information such as current supplier communications, contract negotiations, market conditions, or operational priorities.

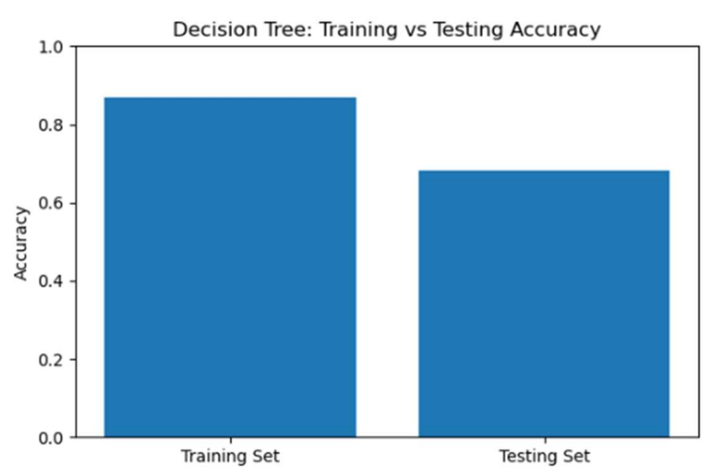
Graphical Charts / Representation

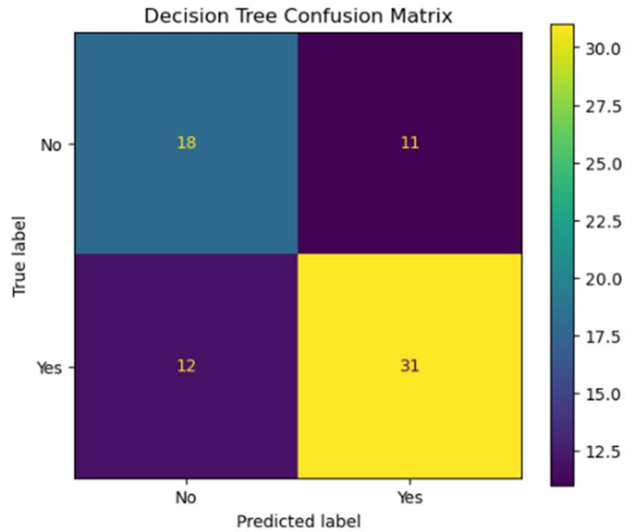


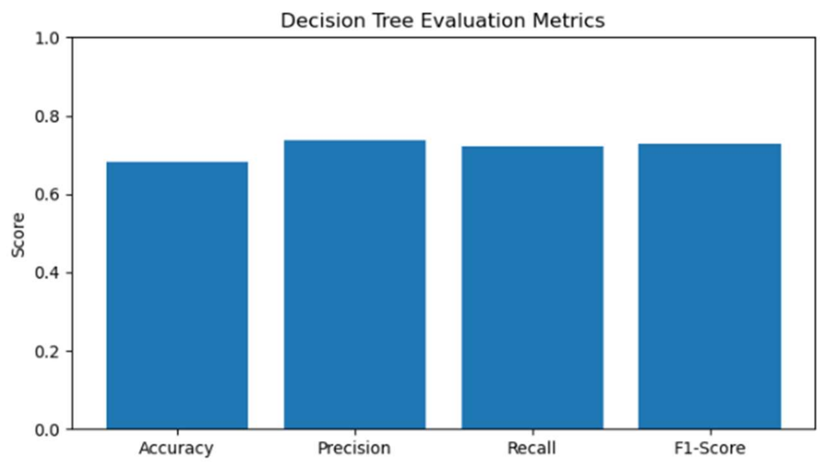



Training Set vs Testing Set Size
Number of Records
0
50
100
150
200
250
300
Training Set
Testing Set


Trained Decision Tree Model
Past_On_Time_Rate <= 0.755
gini = 0.482
samples = 288
value = [117, 171]
class = Yes
Order_Value_CAD <= 1513.31
gini = 0.112
samples = 101
value = [6, 95]
class = Yes
Prior_Delays_Last_6M <= 1.5
gini = 0.482
samples = 187
value = [111, 76]
class = No
gini = 0.0
samples = 1
value = [1, 0]
class = No
Distance_KM <= 76.0
gini = 0.095
samples = 100
value = [5, 95]
class = Yes
Past_On_Time_Rate <= 0.865
gini = 0.365
samples = 104
value = [79, 25]
class = No
Past_On_Time_Rate <= 0.965
gini = 0.474
samples = 83
value = [32, 51]
class = Yes
Shipping_Mode_Local Courier <= 0.5
gini = 0.444
samples = 6
value = [2, 4]
class = Yes
Seasonal_Demand_Index <= 0.795
gini = 0.062
samples = 94
value = [3, 91]
class = Yes
Order_Value_CAD <= 4663.93
gini = 0.484
samples = 51
value = [30, 21]
class = No
Seasonal_Demand_Index <= 1.94
gini = 0.14
samples = 53
value = [49, 4]
class = No
Number_of_Line_Items <= 2.5
gini = 0.432
samples = 73
value = [23, 50]
class = Yes
Number_of_Line_Items <= 1.5
gini = 0.18
samples = 10
value = [9, 1]
class = No
gini = 0.0
samples = 4
value = [0, 4]
class = Yes
gini = 0.0
samples = 2
value = [2, 0]
class = No
gini = 0.5
samples = 2
value = [1, 1]
class = No
gini = 0.043
samples = 92
value = [2, 90]
class = Yes
gini = 0.444
samples = 24
value = [8, 16]
class = Yes
gini = 0.302
samples = 27
value = [22, 5]
class = No
gini = 0.077
samples = 50
value = [48, 2]
class = No
gini = 0.444
samples = 3
value = [1, 2]
class = Yes
gini = 0.444
samples = 12
value = [8, 4]
class = No
gini = 0.371
samples = 61
value = [15, 46]
class = Yes
gini = 0.0
samples = 1
value = [0, 1]
class = Yes
gini = 0.0
samples = 9
value = [9, 0]
class = No


Decision Tree: Training vs Testing Accuracy
Accuracy
0.0
0.2
0.4
0.6
0.8
1.0
Training Set
Testing Set


Decision Tree Confusion Matrix
No
Yes
True label
18
11
12
31
No
Yes
Predicted label
30.0
27.5
25.0
22.5
20.0
17.5
15.0
12.5


Decision Tree Evaluation Metrics
Score
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
Precision
Recall
F1-Score